a

## Task

Colour fundus photograph in, four lesion probability maps out

- MA microaneurysm
- HE hemorrhage
- EX hard exudate
- SE soft exudate

## Data collection

7 test sources · 4,923 images

| Training sources | |
|---|---|
| IDRiD | 81 |
| DDR | 757 |
| **Independent sources** | |
| TJDR-std | 257 |
| TJDR-uwf | 304 |
| FGADR | 1,842 |
| Retinal-Lesions | 1,593 |
| DiaRetDB1 | 89 |

## Model collection

38 variants in the registry

- Tier 1 · main table — 17
- Tier 2 · if time allows — 8
- Tier 3 · paper audit only — 13

Reproducibility level R0–R4 recorded for every variant; failures stay in the denominator

## Four evaluations

- **Paper audit** *L1* — numbers as reported, and where they came from
- **Author weights** *E1* — released checkpoints, three numbers each
- **Author recipe** *E1r* — retrained with the authors' own pipeline
- **Unified retraining** *E2* — shared data, budget, augmentation, selection

## Tests and metrics

**Internal test** — official test set of the training source

**Independent test** — sources never used for training

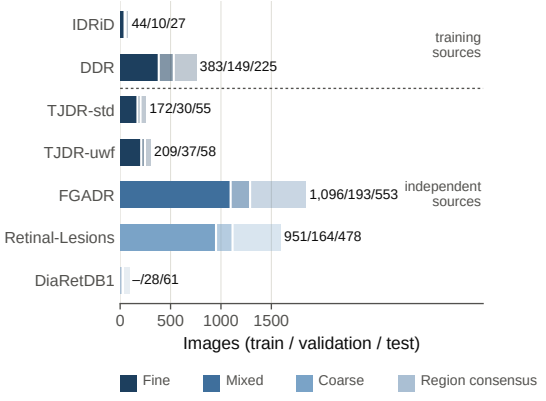
**Primary** — exact pooled AUPR inside the field of view; mAUPR

**Secondary** — Dice, IoU; lesion-level recall by lesion size

**Statistics** — 3 seeds; image bootstrap; Friedman + Nemenyi

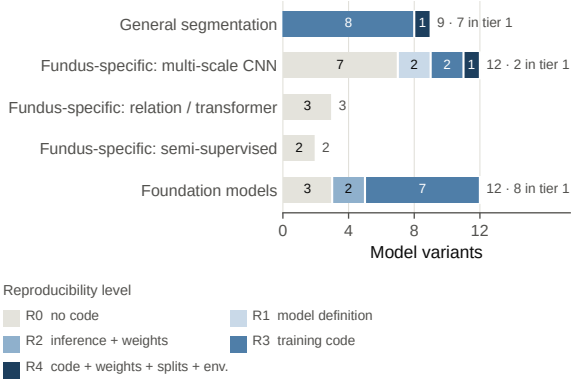
b

## Test sources and frozen splits



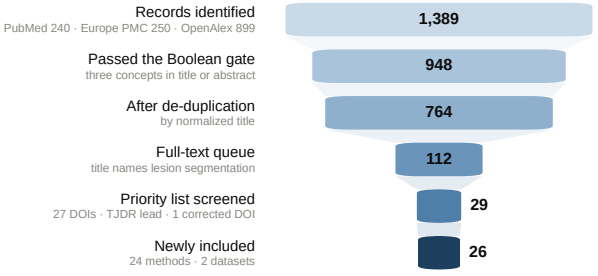

c

## Registered models by family and reproducibility level




d

## Literature search and screening




Fixed search 2015-01-01 to 2026-09-30; with the snowballed corpus the ledger holds 86 records.

e

## Factors assessed

| Training source *E3b* | Label granularity *E4* | Training amount *E8* | Input resolution *E7* |
|---|---|---|---|
| IDRiD · DDR · TJDR-std · FGADR | fine · mixed · coarse, alone and combined | 10–100% of DDR, merged sources | 512 · 1,024 · 1,440 · native patches |
| **Loss** *E9* | **Extra input** *E6* | **Foundation models** *E5* | **Training recipe** *E2b, E2c* |
| CE · Dice · CE + Dice · bin loss | image alone or with vessel map | zero-shot · oracle prompt · PEFT · encoder + decoder | source recipe (B1); learning rate ×⅓, 1, 3 |

One factor changes at a time; everything else stays fixed. E3b–E9 use five representative models: U-Net, nnU-Net, M2MRF, SAM2-UNet, RETFound + UPerNet.